Accessible Quantum Natural Gradient: Optimization with Measurement-Accessible Tangent Geometry

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Natural-gradient methods precondition variational optimization with a state-space information metric, whereas practical quantum learning pipelines act through restricted measurements and low-dimensional classical readouts. We introduce the accessible quantum natural gradient (AQNG), which replaces the full quantum Fisher metric by the pullback of the score geometry visible through a specified measurement interface. For retained commuting readout features, the accessible metric is an exact projected-score Gram matrix, not merely a low-rank numerical approximation to the full QFIM. We evaluate AQNG in a 470-cell benchmark, together with a 15-cell matched large/deep extension containing Adam and SGD. Across 240 generic dataset–architecture cases, cross-fitted aligned AQNG has the lowest pooled mean test loss, but its advantage over random-rank AQNG and first-order baselines is modest and architecture dependent. In the matched large/deep comparison, aligned AQNG outperforms SGD in all 15 cases and Full-QNG in 14 of 15, while its difference from Adam is not statistically resolved. Geometry and trainability also separate sharply: in SU(2)-Haar scaling, normalized aligned retention grows far above the random-rank baseline, yet optimization gains are nonmonotonic; in a number-conserving $U(1)$ control, accessibility remains near complete while the supervised optimization signal collapses by orders of magnitude; and increasing readout order can raise retained tangent mass while worsening optimization through severe metric conditioning. These results support accessible geometry as a controllable and diagnostically interpretable optimization resource, but not as a substitute for task-gradient signal or conditioning.

I. INTRODUCTION

The geometry of parameterized quantum states controls how changes in circuit parameters alter the represented state. For pure states, this local geometry is captured by the Fubini–Study metric, equivalently the quantum Fisher information matrix up to convention, and motivates quantum natural-gradient optimization [1, 2]. This viewpoint is attractive because it replaces Euclidean parameter distance by a state-sensitive notion of distance. It also raises a practical question that is usually left implicit: which part of the state-space geometry is actually visible to the learning interface used by the experiment?

A variational learner rarely observes the full state. It interacts with measurement outcomes, retained observables, marginals, or a small classical feature vector. Consequently, two tangent directions that are distinct in the full quantum metric can become indistinguishable after measurement and readout restriction. Conversely, a low-dimensional readout can be strongly aligned with a small set of task-relevant tangent directions even when it retains only a small fraction of the total state-space information. The distinction between full geometry and accessible geometry therefore matters both statistically and algorithmically.

The present work follows a sequence of increasingly operational geometric questions. First, Fubini–Study metric conditioning was used as a trainability control variable: the Sculpture framework meta-learns circuit initializations that improve the condition number of the full metric [3]. Second, under an isotropic tangent model, fixed-basis readout accessibility was shown to obey a rank law: the expected retained tangent mass is set by the readout rank relative to the score-space dimension [4]. Third, the isotropy assumption was removed by introducing a spectral orientation framework in which retained information depends jointly on the rank baseline, the tangent-covariance spectrum, and the relative orientation of the readout subspace [5]. The current paper closes the algorithmic loop: instead of using accessibility only as a diagnostic, we use the measurement-accessible tangent metric itself to precondition optimization.

There is related work on measurement-limited feature visibility from a different direction. Gross and Rieser formulate quantum extreme learning machines through Pauli-transfer matrices and characterize which encoded Pauli features are decodable through the row space selected by the reservoir dynamics and measurement operators [6]. Their problem is representational: a fixed quantum reservoir transforms an input feature library and a classical readout learns from the resulting features. Our problem is differential: the object being projected is the parameter-score tangent of a trainable circuit, and the projected geometry is subsequently used as an optimization metric. The common linear-algebraic skeleton is a projection onto an observable subspace, but the tangent object, normalization, learning dynamics, and operational question are different.

We introduce the *accessible quantum natural gradient* (AQNG). For a commuting retained readout, AQNG builds a covariance-normalized score Jacobian and uses its pullback Gram matrix as the preconditioner. This con-

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struction admits an exact projected-score interpretation: the accessible metric is the Gram matrix of the component of the full classical score that lies in the retained feature span. It is therefore not defined by approximating the full QFIM in matrix norm. Instead, the metric is tied to a physically specified measurement interface.

The empirical study is designed to separate three questions that are often conflated: whether a tangent direction is visible to the measurement, whether the resulting accessible metric is numerically well conditioned, and whether the supervised objective still carries a usable gradient signal. We use rank-matched physical, cross-fitted aligned, and random readouts; compare against full QNG, block QNG, Adam, and SGD; test analytic and finite-shot regimes; and include a number-conserving $U(1)$ control in which accessibility and trainability separate sharply. The central conclusion is deliberately narrower than a universal optimizer claim: accessible geometry can be a useful and interpretable optimization resource, but retained tangent information alone does not determine trainability.

The main contributions are:

1. an exact accessible-metric construction for retained commuting readouts, together with normalization, damping, and trust controls suitable for optimization;
2. a rank-matched experimental design that separates physical orientation, cross-fitted tangent alignment, and random-subspace baselines without changing readout rank;
3. a fixed-protocol multi-dataset benchmark and scaling study showing that AQNG can outperform full-QNG and SGD in specific regimes while remaining competitive rather than uniformly superior to Adam;
4. a diagnostic decomposition of optimization failure into measurement accessibility, metric conditioning, and task-gradient signal, illustrated by the readout-order and $U(1)$ controls.

II. MEASUREMENT-ACCESSIBLE TANGENT GEOMETRY

A. Score-space representation

Consider a parameterized state measured by a fixed commuting measurement with outcome probabilities $p_a(\boldsymbol{\theta})$ on a known support Ω. For outcome $a \in \Omega$, define the classical score

$$s_{ai} = \partial_i \log p_a, \tag{1}$$

and the probability-weighted score matrix

$$S_{ai} = \sqrt{p_a}\, s_{ai}. \tag{2}$$

By the definition of classical Fisher information,

$$G_{\mathrm{C}} = \sum_a p_a s_a s_a^{\mathsf{T}} = S^{\mathsf{T}} S. \tag{3}$$

Moreover, differentiating $\sum_a p_a = 1$ gives $\sum_a p_a s_{ai} = \sum_a \partial_i p_a = 0$, so every score vector lies in the centered outcome subspace. For any fixed quantum measurement, the associated CFIM is bounded by the QFIM in the same convention, $G_{\mathrm{C}} \preceq G_{\mathrm{Q}}$ [1].

Now retain r commuting readout functions $f_1, \dots, f_r$ of the same outcomes. Let $F \in \mathbb{R}^{|\Omega| \times r}$ contain these functions and let

$$\widetilde{F}_{a\alpha} = F_{a\alpha} - \sum_b p_b F_{b\alpha} \tag{4}$$

be their centered version. Introduce

$$B = D_p^{1/2} \widetilde{F}, \qquad \Sigma = B^{\mathsf{T}} B, \tag{5}$$

where $D_p = \mathrm{diag}(p)$. Direct differentiation of the readout expectations gives

$$\begin{aligned} J_{\alpha i} = \partial_i \sum_a p_a F_{a\alpha} &= \sum_a p_a s_{ai} F_{a\alpha} && (6)\\ &= \sum_a p_a s_{ai} \widetilde{F}_{a\alpha}, && (7) \end{aligned}$$

where the last equality uses the centered-score identity above. Hence

$$J = \partial_{\boldsymbol{\theta}} \langle F \rangle = B^{\mathsf{T}} S. \tag{8}$$

AQNG uses the covariance-normalized pullback

$$\boxed{G_{\mathrm{acc}} = J^{\mathsf{T}} \Sigma^{+} J.} \tag{9}$$

B. Exact projection identity

Equation (9) has a direct geometric interpretation. Substituting $J = B^{\mathsf{T}} S$ and $\Sigma = B^{\mathsf{T}} B$ gives

$$\begin{aligned} G_{\mathrm{acc}} &= S^{\mathsf{T}} B (B^{\mathsf{T}} B)^{+} B^{\mathsf{T}} S && (10)\\ &= S^{\mathsf{T}} P_B S, && (11) \end{aligned}$$

where $P_B = B(B^{\mathsf{T}} B)^{+} B^{\mathsf{T}}$ is the orthogonal projector onto the probability-weighted centered readout span $\mathrm{col}(B)$. Since $0 \preceq P_B \preceq I$,

$$0 \preceq G_{\mathrm{acc}} \preceq G_{\mathrm{C}} \preceq G_{\mathrm{Q}}, \tag{12}$$

with the last inequality supplied by the quantum Fisher bound above [1]. Thus, for commuting retained features, the accessible metric is exactly the Gram matrix of the projected measurement score. It is not defined as a generic low-rank approximation to G_{Q}; it is the metric induced by a specified measurement interface. Figure 1 summarizes the construction.

The covariance normalization removes arbitrary basis scaling inside a fixed retained span. Indeed, replacing B by BM for any nonsingular M leaves $\mathrm{col}(B)$ unchanged, hence leaves the orthogonal projector P_B and therefore G_{acc} unchanged by Eq. (11).

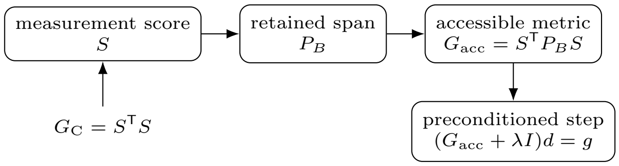

FIG. 1. Accessible natural-gradient construction. The retained readout span projects the measurement score before the pullback metric is formed; the resulting metric preconditions the task gradient.

C. Retention and orientation

To diagnose which score directions a rank-r readout captures, we use the normalized tangent-score covariance introduced in Refs. [4, 5], $C \succeq 0$ with $\operatorname{Tr} C = 1$. If P projects onto the retained score subspace, the retained mass and rank-normalized retention are

$$R = \operatorname{Tr}(PC), \qquad \rho = \frac{R}{r/N}, \tag{13}$$

where N is the centered score-space dimension. Under a uniformly random rank-r subspace, $\mathbb{E}\rho = 1$ [4, 5]. Values $\rho \gg 1$ therefore diagnose orientation beyond the rank baseline. The corresponding random-subspace variance is controlled by the spectrum of C, which separates readout rank, tangent anisotropy, and relative orientation [5].

D. Scope

Equation (11) is an operational statement only when the retained functions are jointly estimable from the same measurement outcome record. In this work that condition is enforced by using diagonal functions of computational-basis outcomes. If one instead wishes to combine incompatible observables, the measurement protocol and its joint classical record must first be specified; only then can an analogous score-space construction be defined.

III. ACCESSIBLE QUANTUM NATURAL GRADIENT

Given a supervised loss $L(\boldsymbol{\theta})$ with Euclidean gradient $g = \nabla_{\boldsymbol{\theta}} L$, AQNG computes a damped preconditioned direction

$$d_{\mathrm{A}} = (\widehat{G}_{\mathrm{acc}} + \lambda I)^{-1} g, \qquad \boldsymbol{\theta} \leftarrow \boldsymbol{\theta} - \eta d_{\mathrm{A}}. \tag{14}$$

The full-QNG control uses the full quantum metric instead, following quantum natural-gradient optimization [2]. From Eq. (11), $G_{\mathrm{acc}} = (P_B S)^{\mathsf{T}}(P_B S)$, so the accessible metric admits the explicit Gram factorization $A^{\mathsf{T}} A$ with $A = P_B S$. This factorization permits primal, dual, or SVD-based linear solves depending on parameter and retained-score dimensions. Metric evaluations are cached for a fixed number of optimization steps as an implementation choice.

A. Metric normalization and damping

Metric scale and damping cannot be varied independently: replacing G by cG while holding λ fixed changes $(cG + \lambda I)^{-1}$ in a direction-dependent way unless λ is rescaled with c. The primary configuration therefore normalizes each metric by its trace before applying a common absolute damping coefficient. Additional controls use no normalization or maximum-eigenvalue normalization and alternative damping conventions based on the mean or maximum eigenvalue. We also impose a Euclidean direction cap and an accessible-metric trust radius. These controls bound the accepted update when the estimated metric is nearly singular; consequently, update magnitude alone is not treated as evidence that a vanishing-gradient problem has been solved.

B. Rank-matched readout designs

The primary readout comparison keeps rank fixed and changes only orientation. The physical readout is the local computational-basis Z span. A random control draws a subspace of exactly the same rank from the centered outcome-function space. The aligned control is fit from an independent calibration set of tangent scores: the leading rank-matched score directions are estimated on calibration tangents and then frozen before optimization and evaluation. Labels are not used in this calibration step. This cross-fit construction avoids evaluating an aligned subspace on the same tangent sample used to fit it.

The rank-matched comparison is important because increasing the number of retained observables changes both dimension and orientation. By holding rank fixed, physical, random, and aligned AQNG test whether orientation relative to the tangent covariance changes optimization at a fixed readout dimension. A separate readout-order ablation deliberately changes rank by moving from weight-one diagonal Pauli functions to all diagonal strings through weight two.

C. Finite-shot estimation

In finite-shot mode, all retained diagonal features are derived from the same computational-basis bitstrings. Known outcome support is fixed structurally rather than inferred from observed nonzero counts. For the number-conserving control, the support is the corresponding fixed-Hamming-weight sector. A symmetric Dirichlet pseudocount regularizes empirical probabilities on that known support before constructing score and covariance estimates. This prevents unobserved-but-allowed outcomes

Pseudo-algorithm: Accessible quantum natural gradient (AQNG)

Input: initial parameters $\boldsymbol{\theta}_0$, learning rate η, damping λ, metric refresh period K, supervised minibatches $\{\mathcal{B}_t\}$, metric minibatches $\{\mathcal{M}_t\}$, commuting readout family F, optional calibration tangents $\mathcal{C}$.

1. Optional calibration. If an aligned readout is used, estimate a rank-matched retained subspace from the calibration tangents $\mathcal{C}$ and freeze it before supervised training.

2. For $t = 0, 1, \dots, T-1$**:**

1. Estimate the supervised gradient $g_t = \nabla_{\boldsymbol{\theta}} L_{\mathcal{B}_t}(\boldsymbol{\theta}_t)$.
2. If $t = 0$ or $t \bmod K = 0$, estimate outcome probabilities and score vectors on $\mathcal{M}_t$; build the centered readout matrix B_t; compute $J_t = B_t^{\mathsf{T}} S_t$ and $\Sigma_t = B_t^{\mathsf{T}} B_t$; then form the accessible metric $\widehat{G}_{\mathrm{acc},t} = J_t^{\mathsf{T}} \Sigma_t^{+} J_t$.
3. Normalize $\widehat{G}_{\mathrm{acc},t}$ and apply the damping and trust-region controls.
4. Solve $d_t = (\widehat{G}_{\mathrm{acc},t} + \lambda I)^{-1} g_t$.
5. Clip d_t if the Euclidean step norm or the metric trust radius exceeds its cap.
6. Update $\boldsymbol{\theta}_{t+1} = \boldsymbol{\theta}_t - \eta d_t$.

3. Output: final parameters $\boldsymbol{\theta}_T$.

FIG. 2. Pseudo-algorithm for AQNG. The metric can be refreshed less often than the loss gradient and can use an independently sampled metric minibatch.

from producing artificial support collapse at finite shot count.

The finite-shot Full-QNG control uses an analytic full metric while its loss is sampled. It is therefore an optimizer reference, not a hardware-resource-matched baseline. Shot accounting for AQNG reports calibration and training resources separately.

IV. EXPERIMENTAL PROTOCOL

A. Benchmark matrix

The benchmark comprises 470 experimental conditions and 3160 completed optimizer evaluations. The primary benchmark is the Cartesian product of four binary datasets (Iris, Breast Cancer, Wine, and binary Digits), four six-qubit circuit families (RyRz–CZ, SU(2)–CNOT, SU(2)–Haar, and a number-conserving $U(1)$ family), and 20 paired stochastic seeds, giving 320 dataset–architecture cells. Each primary cell contains eight optimizers: physical-, random-, and aligned-readout AQNG; an AQNG-Z_0 minimal-readout control; full QNG; block QNG; SGD; and Adam.

The auxiliary matrix contains 40 qubit-scaling cells, 30 depth-scaling cells, 40 readout-order cells, and 40 finite-shot cells. Qubit scaling uses SU(2)–Haar and $U(1)$ circuits at $n \in \{4, 6, 8, 10\}$ with five seeds. Depth scaling uses the same two families at depths $L \in \{1, 3, 5\}$ with five seeds. The readout-order ablation uses Iris and Digits with SU(2)–Haar and $U(1)$, comparing weight-one diagonal readouts with the complete diagonal Pauli span through weight two. Finite-shot experiments use the same two datasets and two circuit families at 1000 and 10000 shots.

A separate large/deep extension supplies Adam and SGD at $(n, L) = (8, 3), (10, 3), (6, 5)$ for SU(2)–Haar, using the same five paired seeds as the corresponding AQNG cells. These 15 conditions were added after the main benchmark had been analyzed and are therefore kept outside the primary hypothesis family.

B. Shared training controls

Within a paired comparison, methods share the same dataset split, circuit family, initialization seed, training budget, and loss definition. The reference runs use 20 optimization steps, a loss minibatch of four samples, a metric minibatch of two samples, and a metric refresh every two steps. The default rank-matched orientation experiment uses a weight-one diagonal readout. Cross-fitted aligned readouts are calibrated on independent tangent samples and then frozen before supervised training.

AQNG orientation methods use learning rate 0.06 and damping 0.003 under trace metric normalization and absolute damping. Baseline learning rates were selected in a restricted tuning stage using only Iris/SU(2)–Haar seed 0: 0.10 for SGD and 0.06 for Adam; AQNG-Z_0 and block-QNG use the same 0.06/0.003 pair. The tuning stage excludes the $U(1)$ control and does not inspect test outcomes from the main benchmark.

C. Statistical reporting

The unit of pairing is the stochastic seed within a fixed dataset–architecture–protocol cell. Primary method comparisons use paired Wilcoxon signed-rank tests and Holm correction over the planned family of comparisons. Pooled means are reported as descriptive aggregates unless a corresponding paired test is defined. The large/deep extension is analyzed separately, with both setting-wise

TABLE I. Pooled generic primary benchmark over 240 cases. Lower test loss and higher accuracy are better.

Method	Test loss	Accuracy
AQNG-aligned	**0.5945**	**0.7947**
AQNG-random	0.5956	0.7914
SGD	0.6084	0.7858
Adam	0.6203	0.7731
AQNG-physical	0.6229	0.7878
AQNG-Z_0	0.6352	0.7828
Full-QNG	0.6562	0.7689
Block-QNG	0.6781	0.7606

five-seed comparisons and a pooled 15-case paired summary.

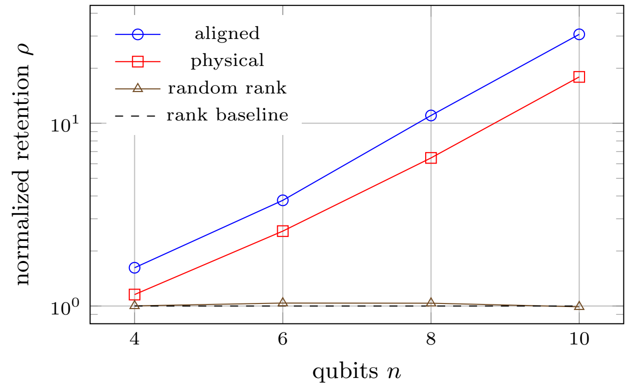

FIG. 3. SU(2)–Haar qubit scaling. The random rank-matched control remains close to $\rho = 1$, whereas the structured readouts become increasingly super-baseline with system size.

V. RESULTS

A. Generic primary benchmark

We first pool the three nonconserving circuit families across the four datasets and 20 seeds, giving 240 paired cases per method. Table I reports descriptive pooled means. Cross-fitted aligned AQNG has the lowest mean test loss and the highest mean test accuracy, but the margin over random-rank AQNG is small. The result therefore supports useful accessible preconditioning without supporting a universal claim that tangent alignment is always the best readout design.

Cellwise outcomes remain architecture and dataset dependent, and the gap between aligned AQNG, random AQNG, SGD, and Adam is much smaller than the gap to full- and block-QNG in this common protocol. The main conclusion from the primary matrix is comparative rather than absolute: an accessible metric can preserve competitive optimization performance while exposing which measurement subspace supplies the preconditioning geometry. Dataset-resolved terminal loss and accuracy curves are reported in Appendix C, Figs. 6–9.

B. Rank baseline and qubit scaling

The scaling experiment gives a clean geometric check of the readout-rank theory. For SU(2)–Haar circuits, the rank baseline r/N decreases rapidly with n, while a random rank-matched readout stays near $\rho = 1$. In contrast, both the physical and cross-fitted aligned readouts become increasingly super-baseline. As shown in Fig. 3, aligned normalized retention grows from approximately 1.62 at $n = 4$ to 30.66 at $n = 10$.

This strong geometric orientation does not translate into a monotonic optimization advantage. The aligned mean test losses at $n = 4, 6, 8, 10$ are 0.221, 0.237, 0.168, and 0.254, respectively. The corresponding physical-readout losses are 0.207, 0.273, 0.238, and 0.241. Thus aligned AQNG is best at $n = 6$ and $n = 8$, but not at $n = 4$ or $n = 10$. Accessibility can therefore increase relative to the null baseline while task utility remains non-monotonic. The corresponding terminal loss and accuracy panels are shown in Appendix C, Figs. 10 and 11.

TABLE II. Mean test loss in the matched large/deep comparison.

Method	$n = 8, L = 3$	$n = 10, L = 3$	$n = 6, L = 5$
AQNG-aligned	**0.168**	0.254	0.139
AQNG-random	0.210	0.252	**0.127**
AQNG-physical	0.238	**0.241**	0.183
Adam	0.231	0.265	0.184
SGD	0.425	0.407	0.367
Full-QNG	0.476	0.430	0.388

C. Matched large/deep comparison with Adam and SGD

The large/deep extension is a post-hoc exploratory comparison that supplies first-order baselines at $(n, L) = (8, 3), (10, 3), (6, 5)$ for SU(2)–Haar on Iris. Table II combines the matched five-seed means. The pooled 15-case loss is 0.1867 for aligned AQNG, compared with 0.1961 for random AQNG, 0.2207 for physical AQNG, 0.2268 for Adam, 0.3998 for SGD, and 0.4312 for full QNG.

In the pooled paired analysis, aligned AQNG beats SGD in 15/15 cases and full QNG in 14/15. The Holm-adjusted Wilcoxon values are $p_{\mathrm{Holm}} = 1.22 \times 10^{-3}$ against SGD and 5.80×10^{-3} against full QNG. Against Adam, aligned AQNG wins 10/15 cases with a mean loss advantage of 0.0400, but the Holm-adjusted test is not significant. Within this exploratory extension, aligned AQNG has significantly lower paired loss than SGD and full QNG after Holm correction, while the difference from Adam is not statistically resolved. Appendix C, Fig. 12, shows both terminal loss and accuracy for the three settings.

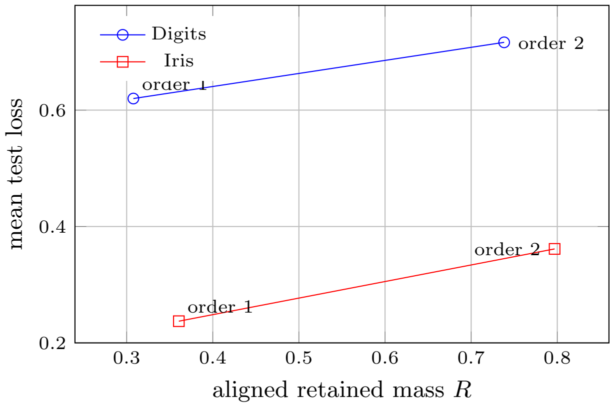

FIG. 4. Readout-order tradeoff for aligned AQNG on SU(2)–Haar. Moving from weight-one to weight-two diagonal readouts increases retained tangent mass but also increases mean test loss on both datasets.

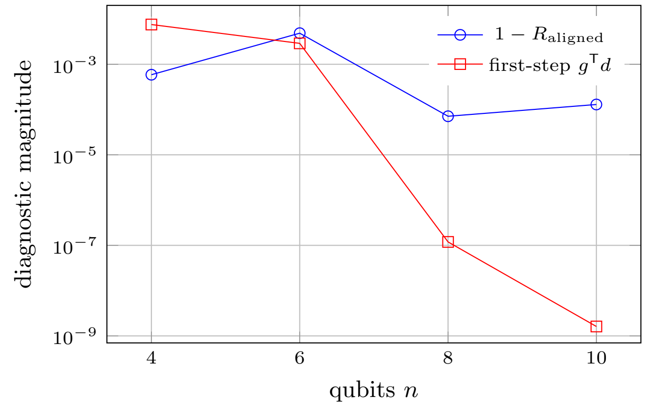

FIG. 5. Diagnostic separation in the number-conserving $U(1)$ scaling control. The aligned readout remains nearly complete, while the first-step descent signal collapses by several orders of magnitude.

D. More retained geometry can hurt

The readout-order ablation provides a direct counterexample to the idea that maximizing retained tangent mass is sufficient. On Iris/SU(2)–Haar, increasing the diagonal readout from weight one to weight two raises aligned retention from 0.361 to 0.797, yet mean test loss worsens from 0.237 to 0.361. On Digits/SU(2)–Haar, aligned retention rises from 0.308 to 0.738, while loss worsens from 0.620 to 0.716. Figure 4 shows the same tradeoff directly. The first-step quantity $g^\top d$ simultaneously drops from 7.05 to 2.57 on Iris and from 3.82 to 1.71 on Digits. Metric diagnostics show that the richer readout is substantially harder to condition. These results motivate condition-aware rather than retention-maximizing readout design.

E. Accessibility does not imply trainability

The number-conserving $U(1)$ family is the strongest negative control. Its physical and aligned readouts remain nearly maximally accessible as system size grows: at $n = 8$, aligned retention is 0.99993, and at $n = 10$ it is 0.99987. Nevertheless, all tested optimizers approach the same chance-like accuracy of 0.4667 at $n = 8$ and $n = 10$, with terminal loss near 2.1333. At the same time, the recorded first-step descent signal collapses by many orders of magnitude. Figure 5 contrasts the small accessibility deficit $1 - R_{\text{aligned}}$ with the rapidly vanishing first-step $g^\top d$. The failure is therefore not measurement blindness; the supervised optimization signal itself becomes too weak to exploit.

This regime is consistent with the gradient-concentration phenomenology associated with barren plateaus [7], but we do not claim a formal barren-plateau scaling theorem: the scaling study uses five seeds and was not designed to estimate an exponential gradient-variance law over a large initialization ensemble. The directly observed conclusion is narrower: near-complete measurement accessibility does not rescue a vanishing task-signal regime.

F. Finite-shot robustness

At 1000 and 10000 shots, AQNG remains operational across the tested Iris and Digits SU(2)–Haar cells. For example, on Digits the aligned losses are 0.661 and 0.624 at 1000 and 10000 shots, respectively, compared with 0.829 and 0.790 for the full-QNG control. On Iris, random-rank AQNG is the best of the tested AQNG variants at both shot counts. The ordering is therefore not monotonic in shot count or readout orientation. Because the finite-shot full-QNG implementation uses an analytic metric oracle, these numbers are optimization references rather than a resource-fair hardware comparison.

VI. GEOMETRIC INTERPRETABILITY

AQNG exposes diagnostics that are hidden by a purely Euclidean optimizer. The relevant notion of interpretability is geometric and mechanistic rather than semantic: the metric identifies which parameter-score directions are visible to the retained measurement interface, how strongly those directions are weighted, and whether the resulting preconditioner is numerically usable.

Three quantities separate distinct failure modes. First, retention R and normalized retention ρ diagnose *accessibility*: whether the readout sees the tangent covariance beyond its rank baseline. Second, the spectrum of G_{acc} diagnoses *conditioning*: whether the visible geometry can be inverted stably after normalization and damping. Third, the raw task-gradient norm or finite-shot signal-to-noise ratio diagnoses *supervised signal*: whether the objective supplies enough directional information for any preconditioner to exploit.

The experiments instantiate all three regimes. SU(2)–Haar scaling produces strongly super-baseline accessible subspaces and competitive optimization. The weight-two readout ablation has substantially higher retention but worse optimization, revealing a conditioning bottleneck. The $U(1)$ scaling control has nearly complete accessibility but a collapsing task signal, revealing a trainability bottleneck that no accessible metric can repair by itself. In this sense, AQNG is not only a preconditioner; it provides a decomposition of why optimization succeeds or fails.

This diagnostic interpretation should not be confused with semantic feature attribution. AQNG does not by itself imply that an eigenvector of the accessible metric corresponds to a human-interpretable concept or to one classical input feature. A semantic layer becomes possible only when the retained observables or tangent directions are tied to a structured feature dictionary with external meaning. The Pauli-transfer framework of Gross and Rieser provides an example of such a representational interpretation for encoded Pauli features in QELMs [6]. Extending AQNG in that direction would require an explicit map from accessible tangent directions to known data features or physically meaningful observables; this is outside the claims of the present study.

VII. DISCUSSION AND CONCLUSION

The experiments support a specific claim: measurement-accessible tangent geometry can be used as a physically grounded preconditioner, and in several regimes it is more useful than the full state-space metric or a Euclidean gradient. They do not support the stronger claim that AQNG is a universally superior optimizer. Random-rank AQNG is often close to aligned AQNG, Adam remains competitive in the large/deep comparison, and the best readout orientation depends on architecture, depth, and system size.

The distinction between *accessibility* and *trainability* is central. A readout can capture nearly all of the relevant tangent covariance while the supervised gradient is effectively absent, as in the large-n $U(1)$ control. Conversely, a readout can retain more tangent mass while producing a less useful preconditioner because the metric becomes poorly conditioned, as in the weight-two ablation. Retention is therefore a resource descriptor, not an optimization objective by itself. A more complete trainability picture requires at least three ingredients: accessible orientation, stable metric spectrum, and surviving task-gradient signal.

This also clarifies the relation to barren plateaus [7]. AQNG rescales and rotates an observed gradient; it cannot create supervised information that has already fallen below the estimation noise floor. In an exact arithmetic model, an inverse metric can compensate some common scaling between gradient and geometry, but finite damping, small-eigenvalue instability, and metric-estimation noise limit that mechanism. Demonstrating barren-plateau mitigation would require a dedicated scaling study of gradient variance or update signal-to-noise with substantially larger initialization ensembles and unclipped update diagnostics. The present $U(1)$ experiment is therefore a gradient-collapse control rather than a formal barren-plateau theorem.

Several resource caveats are also important. Cross-fitted alignment requires calibration tangents, and those measurements are part of the experimental cost. The finite-shot full-QNG control uses an analytic metric oracle, so it is not a hardware-cost-matched comparator. The primary 20-seed benchmark keeps the dataset split fixed, meaning that it probes initialization and optimization stochasticity rather than split-to-split generalization uncertainty. Finally, the exact projector identity used by AQNG assumes a jointly measurable commuting readout; extending the method to noncommuting observable families requires an explicit measurement-grouping model.

The readout-order ablation suggests a concrete next algorithmic step. Rather than choosing the subspace that maximizes retained tangent mass, one can optimize a regularized criterion that trades accessibility against spectral conditioning, for example through eigenvalue truncation, adaptive damping, or a condition-number penalty. A second direction is semantic structure: if the retained observables are tied to known input features or physical quantities, accessible eigendirections may be annotated with domain meaning rather than only geometric meaning. A third is a dedicated trainability-scaling experiment in which the raw gradient, accessible spectrum, metric-estimation noise, and shot complexity are tracked simultaneously.

AQNG converts measurement-accessible tangent geometry from a diagnostic into an optimizer. Its value is not universal dominance over first-order methods; it is the explicit coupling between the preconditioning geometry and the measurement interface. The experiments show that accessibility, conditioning, and task-gradient signal must be assessed separately. When all three are favorable, the accessible metric can provide competitive or improved optimization; when either conditioning or gradient signal fails, high retained tangent mass alone is insufficient.

Appendix A: Implementation details

The reference implementation uses PennyLane 0.45.x with Autograd-compatible circuit derivatives. The accessible metric is formed from retained computational-basis outcome functions and supports primal, dual, and singular-value-based linear solves. In the reference configuration, metrics are trace-normalized, damped with an absolute coefficient $\lambda = 0.003$, refreshed every two optimization steps, and combined with a Euclidean direction cap and an accessible-metric trust radius. These controls are fixed before the main benchmark is evaluated.

The primary physical readout uses local Z functions.

Rank-matched aligned and random controls have exactly the same retained dimension. The aligned basis is estimated from 64 calibration tangents and evaluated on an independent set of 64 held-out tangents before being frozen for training. The order-two ablation replaces the weight-one span by all diagonal Pauli strings through weight two.

For finite-shot probability estimates, the support is fixed from circuit structure. Generic circuits use the full computational-basis support. Number-conserving circuits use the known fixed-Hamming-weight sector. A symmetric Dirichlet pseudocount of $1/2$ is added on the known support before score construction, and the readout covariance receives an additional numerical regularization term. This choice prevents empirical zero counts from changing the model support.

Appendix B: Statistical reporting

Primary hypothesis tests use paired seeds within a fixed dataset–architecture cell. For each planned method comparison, the statistic is computed from paired terminal test-loss differences using the Wilcoxon signed-rank test [8]. The resulting family of p-values is corrected with the sequentially rejective Holm procedure [9]. Means, standard deviations, win counts, and pooled summaries are reported alongside corrected tests because the cellwise optimization outcomes are visibly heterogeneous across circuit families.

The three large/deep settings are analyzed separately from the primary comparison family. They use the same previously selected learning rates and the same paired seeds as the corresponding AQNG cells. We report both the five-seed setting-wise comparisons and a pooled 15-case paired summary.

The qubit-scaling $U(1)$ study is not used to estimate an asymptotic barren-plateau exponent. Barren-plateau claims concern concentration of gradients over initialization ensembles and system-size scaling [7]. Our five-seed control instead establishes the narrower empirical observation reported in the main text: near-complete readout accessibility coexists with a rapidly collapsing supervised optimization signal over the tested sizes. Establishing an exponential gradient-variance law would require a dedicated random-initialization ensemble and explicit scaling fits beyond the present protocol.

Appendix C: Supplementary optimization figures

This appendix reports cell-resolved terminal performance. Each plotted value is the mean over the paired seeds available in that cell; Appendix B describes the statistical analysis. Line plots are used throughout, and labels are suppressed on upper rows to keep the panels readable.

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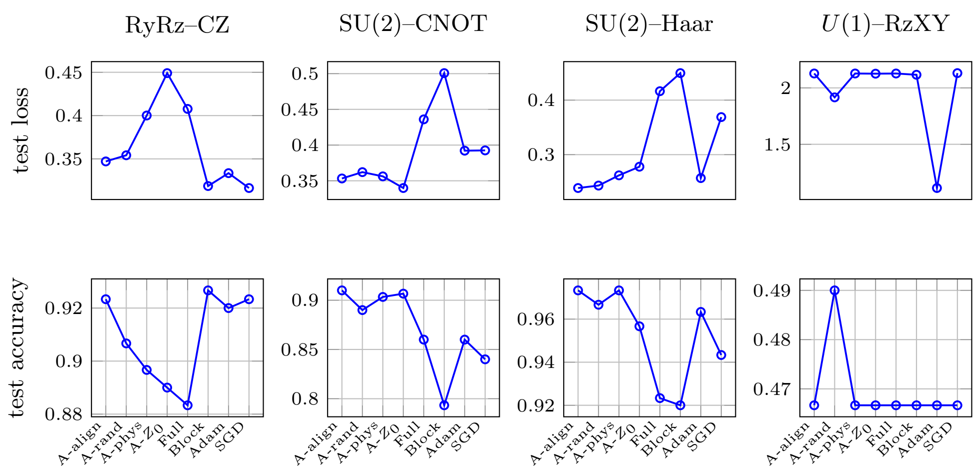

FIG. 6. Iris primary benchmark. Columns are the four circuit families. Mean terminal test loss is shown above and mean terminal test accuracy below. Optimizer labels appear only on the lower row to avoid label overlap.

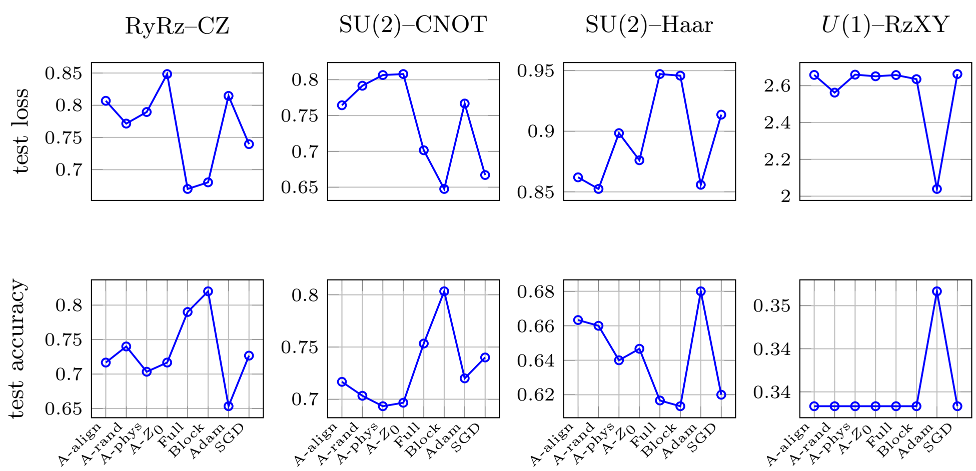

FIG. 7. Breast Cancer primary benchmark. Columns are the four circuit families. Mean terminal test loss is shown above and mean terminal test accuracy below. Optimizer labels appear only on the lower row to avoid label overlap.

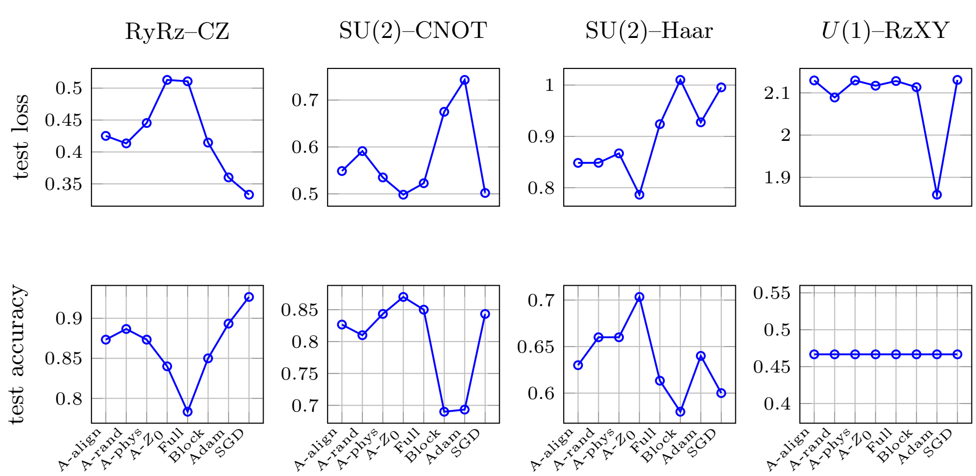

FIG. 8. Wine primary benchmark. Columns are the four circuit families. Mean terminal test loss is shown above and mean terminal test accuracy below. Optimizer labels appear only on the lower row to avoid label overlap.

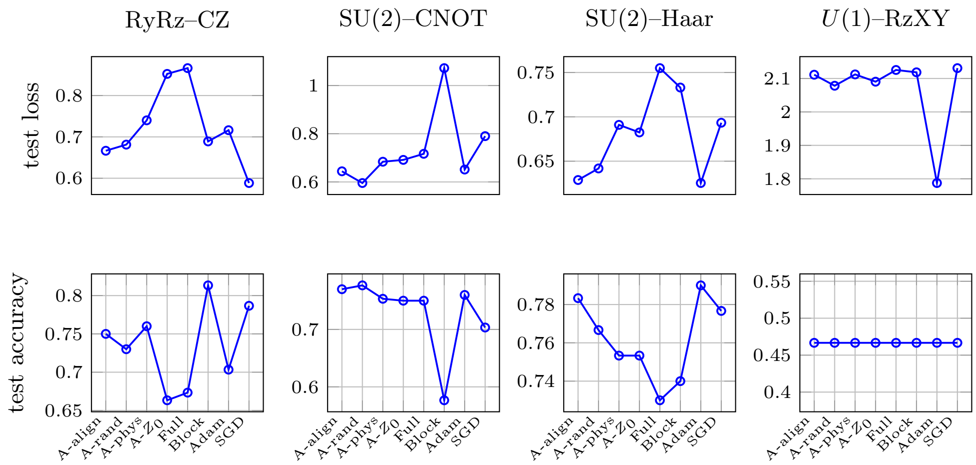

FIG. 9. Digits primary benchmark. Columns are the four circuit families. Mean terminal test loss is shown above and mean terminal test accuracy below. Optimizer labels appear only on the lower row to avoid label overlap.

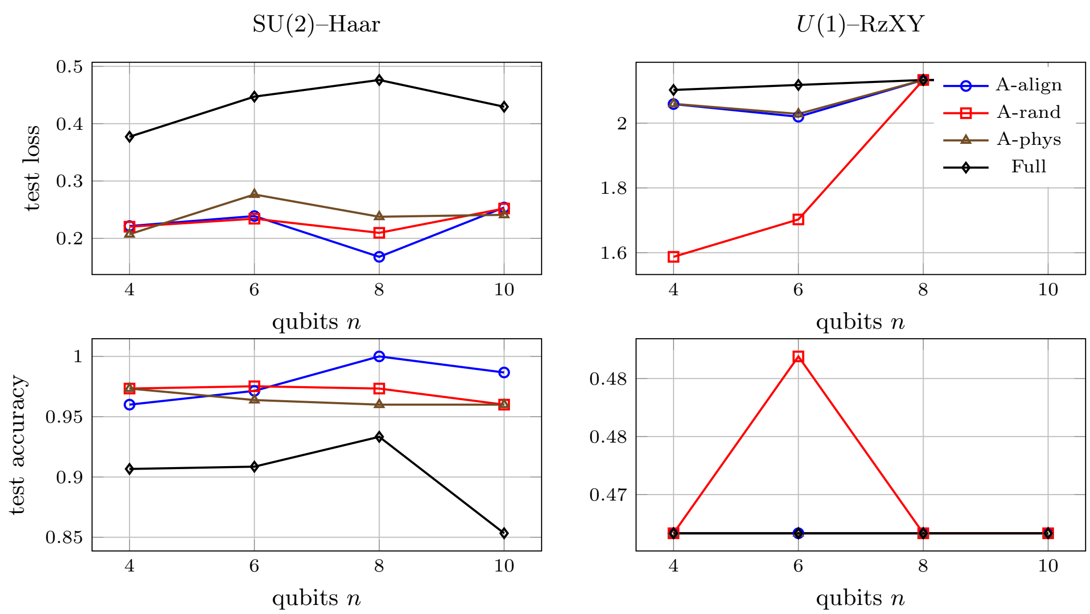

FIG. 10. Scaling cells on Iris at readout order 1. The top row shows mean terminal test loss and the bottom row mean terminal test accuracy. Curves are drawn only where multiple qubit counts were actually evaluated.

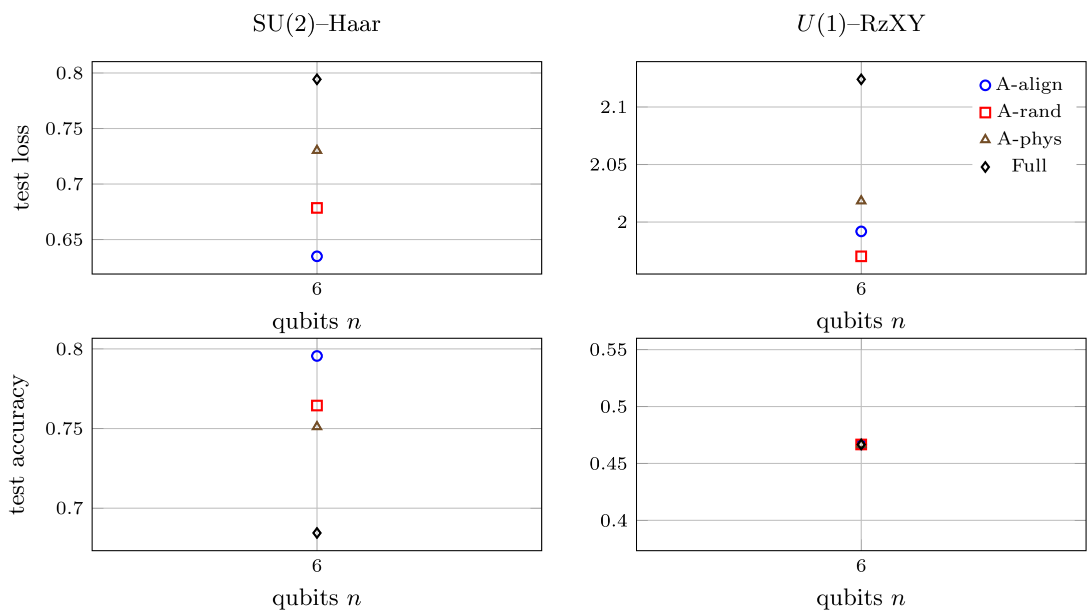

FIG. 11. Scaling cells on Digits at readout order 1. The top row shows mean terminal test loss and the bottom row mean terminal test accuracy. Curves are drawn only where multiple qubit counts were actually evaluated.

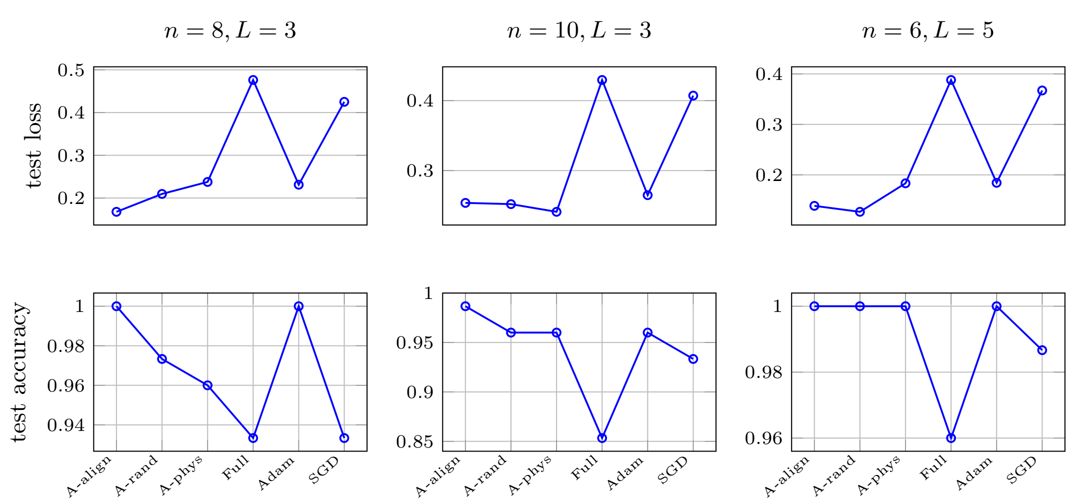

FIG. 12. Matched large/deep follow-up on Iris/SU(2)–Haar. Columns correspond to $(n, L) = (8, 3)$, $(10, 3)$, and $(6, 5)$. The upper row shows mean terminal test loss and the lower row mean terminal test accuracy across the six matched optimizers. This comparison is post-hoc and exploratory.